

**GRAD-E1354: Data Science and Decision Making****1. General information**

|                           |                            |
|---------------------------|----------------------------|
| Course Format             | Onsite                     |
| Instructor(s)             | Dr William Lowe            |
| Instructor's E-mail       | lowe@hertie-school.org     |
| Assistant (if applicable) | behlen@hertie-school.org   |
| Instructor's Office Hours | 2 hours per week, time tbd |

Link to [Study, Examination and Admission Rules and MIA, MDS and MPP Module Handbooks](#)

For information on **course room**, times, and **session dates**, please consult the [Course Schedule](#) on *MyStudies*.

**Instructor Information:**

Dr. Will Lowe is a senior research scientist at the Hertie School. He works on statistical models of text and on the intersection of machine learning and causal inference. His applied research spans legislative politics, political economy, and public policy. He has a PhD in Cognitive Science from the University of Edinburgh and a Bachelor of Arts in Philosophy from the University of Warwick, and has previously held research positions at Harvard University, the University of Nottingham, Trinity College Dublin, the University of Mannheim, and most recently Princeton University. He was for several years a technical architect in a software company and has worked on a range of public-facing academic and civil society projects, including for the ICC's legal tools project.

**2. Course Contents and Learning Objectives**Course contents:

This course is about decision-making in theory and practice. Our theoretical framework will be classical decision theory, informed by the literature on causal inference, machine learning, human and cognitive performance, and bias. Our discussion of practice will be driven by invited speakers, mostly from industry, whose jobs require them to make decisions in a technical context, e.g., to drive sales, run election campaigns, or evaluate government programs.

Please note that all of the course's potential speakers have busy schedules, so the identity of the speaker, order of the sessions, and sometimes the timing of our sessions may be adapted around them. In case of unavoidable time changes, the session will be recorded.

Learning objectives:

The aims of the class is that students have

1. a conceptual understanding of decision theory principles including where it is most, and perhaps least applicable
2. an appreciation of how it can be applied to decisions made by individual humans (and other animals), groups (e.g. bureaucratic and electoral), and machines (e.g. data science and AI).
3. some practical experience with the causal inference tools and treatment targeting strategies

Target group:

There are two target groups for the course: students in the first, primarily but not exclusively MDS students, will be doing the technical implementation of the tools we discuss and would benefit from putting these tools in wider decision-making context, so they better advise decision makers. Students in the second expect to be those decision makers or to be in organizational roles that mediate between data scientists and executives of some kind. The technical elements and more abstract elements of the course material are deliberately presented together to help each group more effectively communicate with the other.

Teaching style:

Lectures, class discussion, and external speakers

Prerequisites:

The course assumes Hertie's Statistics 2, Causal Inference, or their equivalents. Since the probabilities used in decision theory have a statistical but also a causal inference aspect, our prerequisites divide the same way. On the statistical side, we will make regular use of probability theory concepts: conditional and marginal probability distributions, independence, expectation, variance and covariance, and regression functions. Some intuition about generic optimisation will be helpful but not essential. On the causal side, we will assume an understanding of identification, graphical causal models (a.k.a. DAGs), paths, and confounding. Previous experience suggests that without a reasonable grasp of these concepts you will have a hard time following the technical aspects of the lectures. However, this is not a technical or skills course, so if you are willing to ask questions and work through material on your own, this should not stop you. The instructor is also available for questions outside class, independent of formal office hours.

Previous exposure to Bayesian inference, whether in a machine learning, game theoretical, psychological, or epidemiological context, will be very helpful but is not required if you are comfortable with the concepts above.

There are no labs associated with this course, but there is a data analysis assignment that assumes the ability to write code (in a language of your choice). If you are not confident in your abilities here, this assignment can be done as a group.

If you feel you are lacking some prerequisites or are unsure but still wish to take the course and are willing to spend some time preparing, please contact the instructor as soon as you can for suggestions.

Diversity Statement:

Understanding and respect for all cultures and ethnicities is central to the teaching at Hertie. Being mindful of diversity is an important issue for policy professionals in the planning, implementation, and

evaluation of programmes designed for specific groups, populations, or communities. Diversity and cultural awareness will be integrated in the course content whenever possible. Indeed, we will be often point out formal and decision -theoretic motivations of what you may have previously considered to be purely ethical concerns.

### 3. Grading and Assignments

#### Policy on the use of AI tools for assignments

The use of generative AI tools (such as ChatGPT) is **allowed for the final data project**. Class quizzes will be done on paper.

Students must adhere to the specific instructions provided for each assignment. Where AI tools are used, students are expected to clearly indicate how they were used and to ensure the originality and academic integrity of their work. **Students must include a Statement of Authorship in all written assignments.** Improper or undisclosed use of AI tools may be considered a violation of academic integrity policies. Please consult the course instructor in case of doubt. Further guidance is available in the [Artificial Intelligence Tools at the Hertie School - Teaching Guidelines for Instructors and Students](#), including a template for the **Statement of Authorship**, under section 1.1.

#### Composition of Final Grade:

|                                                      |                                       |                    |     |
|------------------------------------------------------|---------------------------------------|--------------------|-----|
| <b>Assignment 1: Quiz</b>                            | In class: week 3                      | Submit via Moodle. | 15% |
| <b>Assignment 2: Quiz</b>                            | In class: week 4                      | Submit via Moodle. | 15% |
| <b>Assignment 3: Quiz</b>                            | In class: week 5                      | Submit via Moodle. | 15% |
| <b>Assignment 4: Quiz</b>                            | In class week 6                       | Submit via Moodle  | 15% |
| <b>Assignment 5: Quiz</b>                            | In class: week 7                      | Submit via Moodle  | 15% |
| <b>Assignment 6: Data analysis / coding exercise</b> | Two weeks after the end of the course | Submit via Moodle  | 20% |
| <b>Participation grade</b>                           |                                       |                    | 5%  |

#### Assignment Details

##### **Quiz Assignments:**

Quizzes happen in the first 10 m of most classes that do not have invited speakers. The weeks listed above may be changed if the speaker needs to be rearranged to a different week.

##### **Data analysis / coding exercise:**

The data analysis coding exercise is a decision problem broken down into a sequence of sub-problems for which you will write up the solution and show your working.

This assignment can but need not be done as part of a group. Students who are not particularly confident in their coding skills will probably want to be part of a group. They should nevertheless be aware that the purpose of the exercise is the analysis and the write up, not the production of code, so there will be plenty of opportunity every member to pull their weight.

One consequence of this is that the quality of the analysis and writeup and the clarity and motivation of the code will gain marks but, unlike some other courses available to MDS students, the elegance, sophistication, production-readiness, or other technical virtues of the code will not.

### **Participation grade**

The participation grade assumes that students take part, not as passive consumers of knowledge, but as active participants in the exchange, production, and critique of ideas—their own ideas and the ideas of others. Therefore, students should come to class not only having read and viewed the materials assigned for that day but also prepared to discuss the readings and to contribute thoughtfully to the conversation. Participation is marked by its active nature, its consistency, and its quality.

**Late submission of assignments:** For each day the assignment is turned in late, the grade will be reduced by 10% (e.g. submission two days after the deadline would result in 20% grade deduction).

**Attendance:** Students are expected to be present and prepared for each class session. Active participation during lectures and seminar discussions is essential. Please note that students can miss up to two sessions (out of twelve) if no course assignments are affected. **Students do not need to inform the course instructor(s) in advance if they are unable to attend a session.** For further information please consult the [Examination Rules](#) §10.

**Academic Integrity:** The Hertie School is committed to the standards of good academic and ethical conduct. Any violation of these standards shall be subject to disciplinary action. Plagiarism, misuse of AI, free riding in group work, and other deceitful actions are not tolerated. See [Examination Rules](#) §16, the Hertie [Plagiarism Policy](#), and [the Hertie Guidelines for Artificial Intelligence Tools](#).

**Compensation for Disadvantages:** If a student furnishes evidence that they are not able to take an examination as required in whole or in part due to disability or permanent illness, the Examination Committee may upon written request approve learning accommodation(s). In this respect, the submission of adequate certificates may be required. See [Examination Rules](#) §14.

**Extenuating circumstances:** An extension can be granted due to extenuating circumstances (i.e., for reasons like illness, personal loss or hardship, or caring duties). In such cases, please contact the course instructor and Examination Office *in advance* of the assignment deadline.

## **4. Course Sessions and Readings**

All course readings can be accessed on the course Moodle page.

Please note that fall semester Moodle course pages are available until 31 March of the following semester. Spring semester Moodle courses are available until 31 September of the following semester. After that, in accordance with German copyright law, they cannot be accessed.

## 1. General Course Readings

In addition to the weekly readings there are several books, available online. These are not intended as textbooks for the course as we will not work through them chapter by chapter.

Note: The readings below are a representative sample of course readings, but individual readings may vary and the moodle pages will be populated with a very wide range of further readings, none of which are required. Data science and to a lesser extent decision theory change quickly and it is more useful to update the readings in real time. Students are welcome to ask the instructor or previous students to get an idea of what the readings have looked like in previous years.

### Session 1: **Decision Theory, the Very Idea**

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|-------------------|-----------------------------------------------------------------|
| Required Readings | Morrow, James (1994) Game Theory for Political Scientists ch. 2 |
| Optional Readings |                                                                 |

### Session 2: **Uncertainty**

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|-------------------|-----------------------------------------------------------------------------|
| Required Readings | Bishop, Christopher M. Pattern recognition and machine learning, 2006. ch.1 |
| Optional Readings |                                                                             |

### Session 3: **Utility**

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|-------------------|----------------------------------------------------------------------------------------|
| Required Readings | McCarty, Nolan, and Adam Meirowitz. Political game theory: An introduction, 2007. ch.4 |
| Optional Readings |                                                                                        |

### Session 4: **Data Science and Decision Making**

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|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Required Readings | C. Fernández-Loría, F. Provost (2022) Causal decision making and causal effect estimation are not the same... and why it matters. INFORMS Journal on Data Science. |
| Optional Readings |                                                                                                                                                                    |

### Session 5: **Decision Making by Humans (and Other Animals)**

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|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Required Readings | - Oaksford, M., & Chater, N. (2009). Précis of Bayesian Rationality: The Probabilistic Approach to Human Reasoning. Behavioral and Brain |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|

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|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | Sciences, 32(01), 69–69.<br>- Gigerenzer, G, and D G Goldstein. "Reasoning the fast and frugal way: Models of bounded rationality," 1996. 103. |
| Optional Readings |                                                                                                                                                |

#### Session 6: **Decision Making by Machines**

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|-------------------|--------------------------------------------------------|
| Required Readings | McElreath, Richard (2016) Statistical Rethinking. ch.1 |
| Optional Readings |                                                        |

#### Session 7: **Decision Making by Groups**

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|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Required Readings | List, Christian, 'Social Choice Theory' in Stanford Encyclopedia of Philosophy <a href="https://plato.stanford.edu/entries/social-choice/">https://plato.stanford.edu/entries/social-choice/</a> |
| Optional Readings |                                                                                                                                                                                                  |

#### Session 8: **Decision Making and Ethics**

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|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| Required Readings | Chouldechova, Alexandra, and Aaron Roth. "A snapshot of the frontiers of fairness in machine learning," 2020. 63 (5). |
| Optional Readings |                                                                                                                       |

#### Session 9: **Practitioner Talk**

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|-------------------|-----------------------------------------------------------------------------------|
| Required Readings | Use the readings from the previous weeks to inform your questions to the speaker. |
| Optional Readings |                                                                                   |

#### Session 10: **Practitioner Talk**

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|-------------------|-----------------------------------------------------------------------------------|
| Required Readings | Use the readings from the previous weeks to inform your questions to the speaker. |
| Optional Readings |                                                                                   |

#### Session 11: **Practitioner Talk**

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|-------------------|-----------------------------------------------------------------------------------|
| Required Readings | Use the readings from the previous weeks to inform your questions to the speaker. |
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| Optional Readings |  |
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### Session 12: **Putting it all Together**

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|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Required Readings | <ul style="list-style-type: none"> <li>- Paul, L. A. and Kieran Healy. Transformative treatments, Nous, 2018</li> <li>- Kohler-Hausmann, I. Eddie Murphy and the dangers of counterfactual causal thinking about detecting racial discrimination. Northwestern University Law Review 2019</li> </ul> |
| Optional Readings |                                                                                                                                                                                                                                                                                                      |